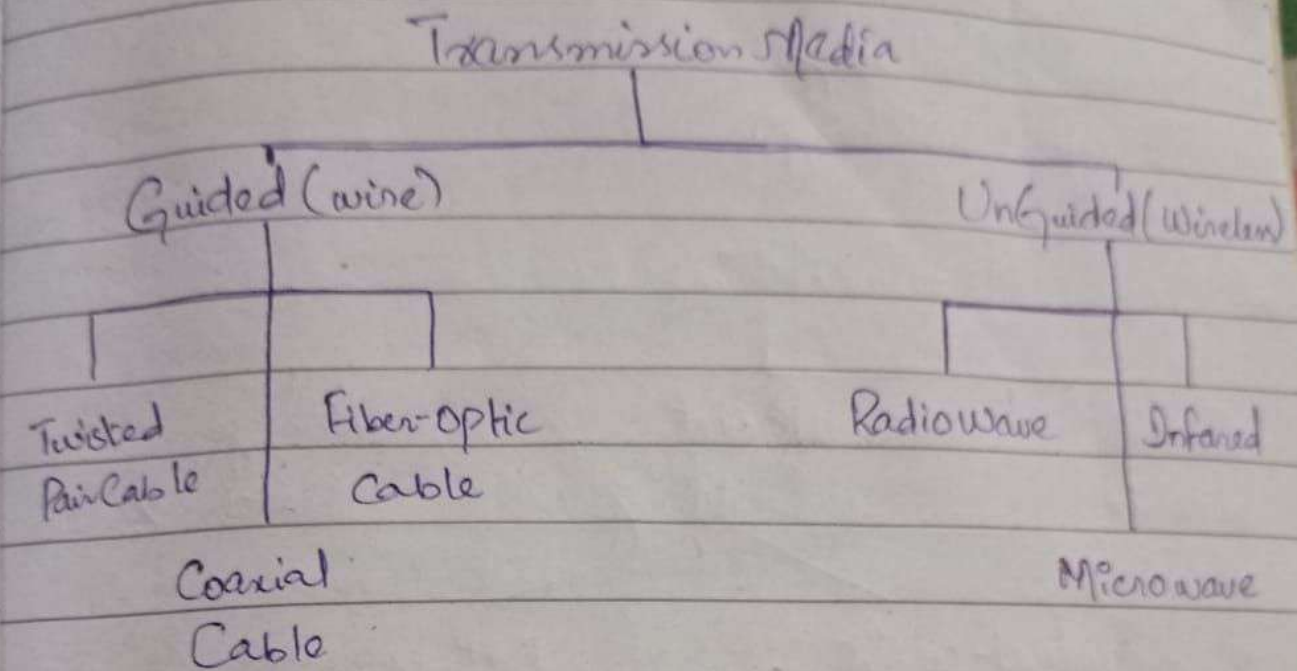


* Classes of Transmission Media.

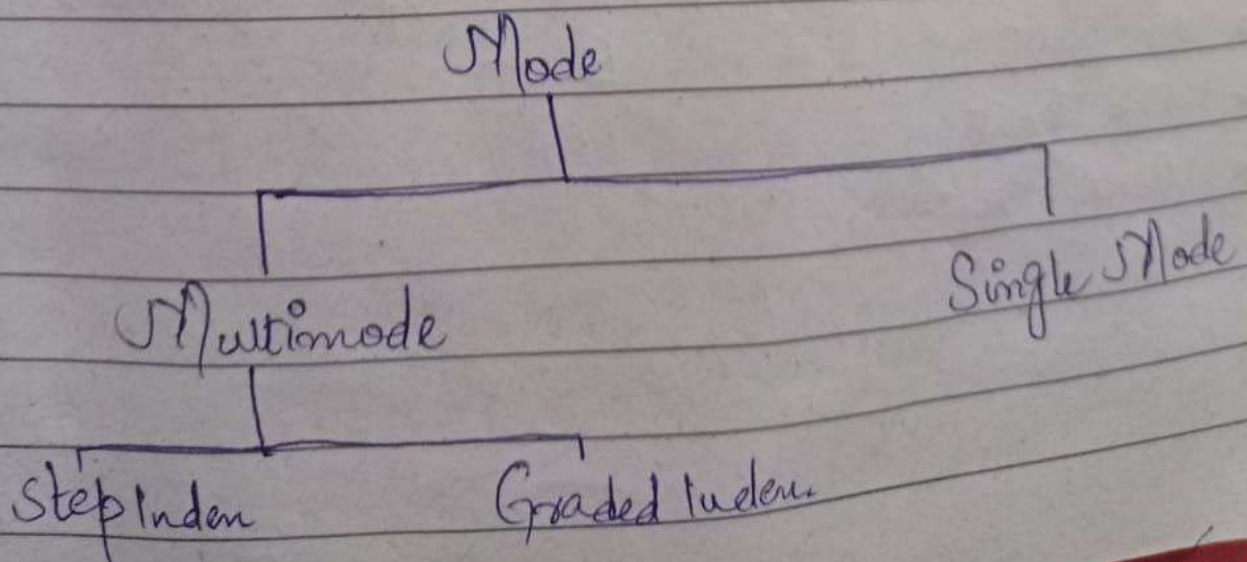


⇒ Twisted Pair Cable

* UTP → Unshielded Twisted Pair Cable

* STP → Shielded Twisted Pair Cable

Propagation Modes.

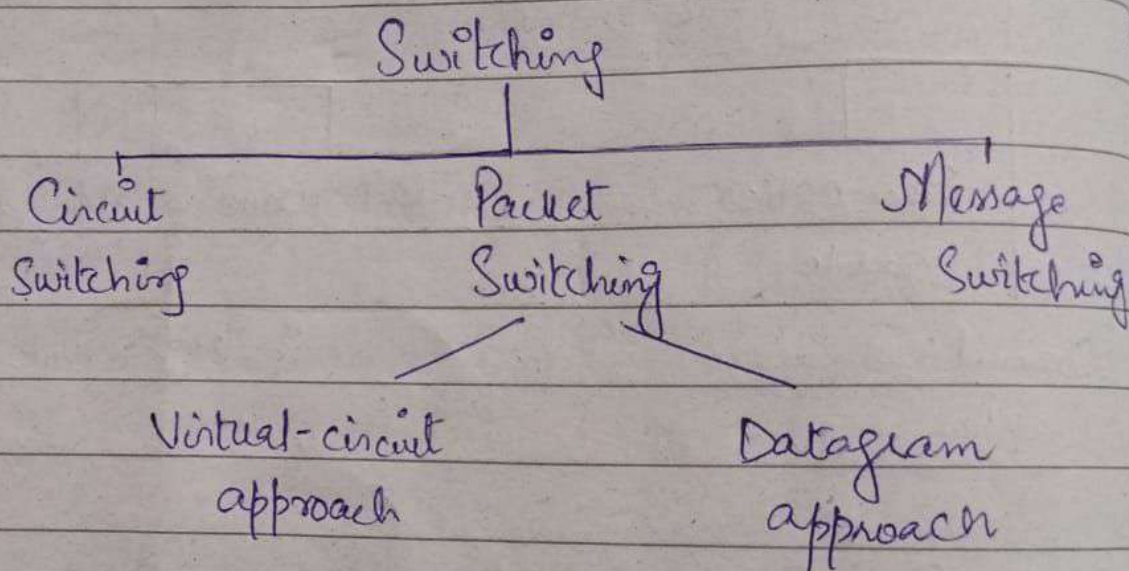


⇒ Three Methods of switching

✓ Circuit Switching.

✓ Packet Switching

✓ Message Switching.



⇒ Two types of lines.

* Point-to-Point line

* Broadcast line.

⇒ Link layer Protocols define three types of addresses

* Unicast

* Multicast

* Broadcast.

⇒ Three sets of PPP Protocols are:

- * Link Control Protocol (LCP)
- * Two Authentication Protocol (APs)
- * Several Network Control Protocols (NCPs).

⇒ Authentication Protocols in PPP

PPP Authentication

Password Authentication
Protocol (PAP)

Challenge Handshake
Authentication Protocol
(CHAP)

→ Media Access Protocol (MAC) Sublayers.

* Random Access Protocol:

✓ ALOHA

✓ CSMA/CD

✓ CSMA/CA

* Controlled Access Protocols:

✓ Reservation

✓ Polling

✓ Token Passing

3rd type on a Net →

⇒ Types of error.

- * Single bit error
- * Burst error.

⇒ Types of coding Scheme-

- * Block coding
- * Convolution Coding

⇒ Circuit Switch Network Requires 3 phases:

- * Connection Setup.
- * Data Transfer
- * Connection Teardown

⇒ Two Sublayers of Data-link layer

- * Data Link Control (DLC).
- * Media Access Control (MAC).

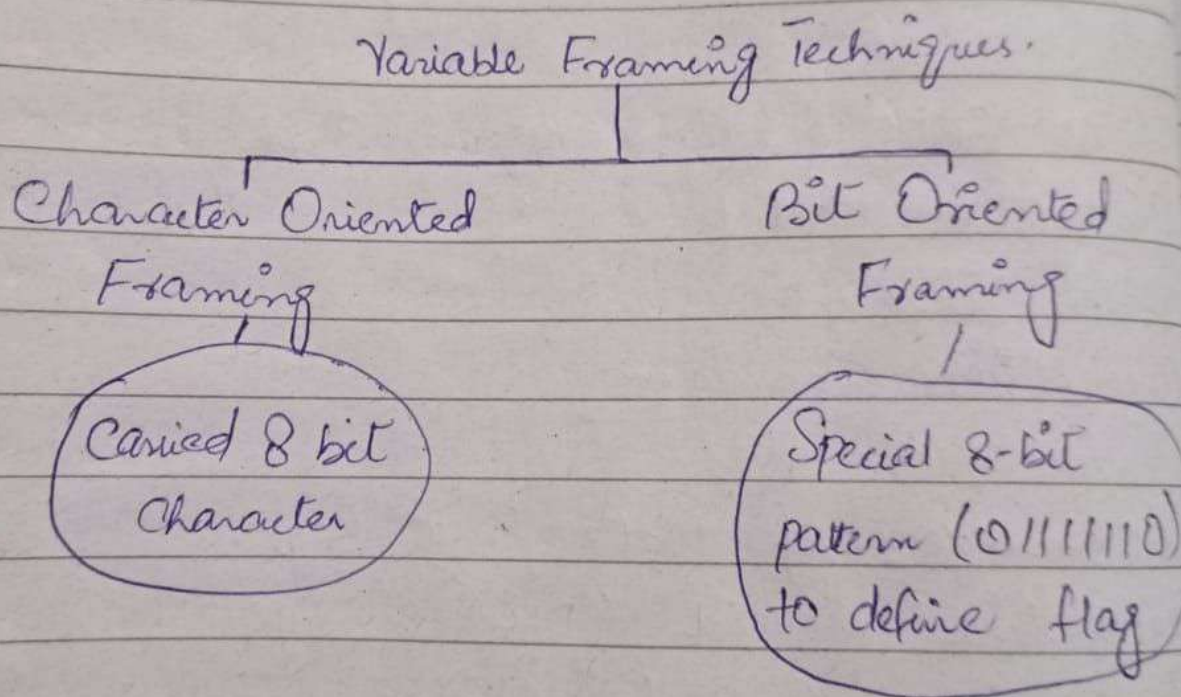
⇒ VIP

- * RJ-45 Male
- * RJ-45 Female.

⇒ DLC Service.

- * Framing
- * Flow Control.
- * Error Control.

⇒ Variable Framing Techniques.



⇒ HDLC defines three types of frames.

- * Information frames (I-frames)
- * Supervisory frames (S-frames)
- * Unnumbered frames (U-frames).

standard
⇒ SONEI' includes four functional layers.

- ✓ The Path layer
 - ✓ The Line layer
 - ✓ The Section layer
 - ✓ The Photonic layer
- } Data link
- } Physical link

⇒ SONEI Networks divided into three categories.

- ✓ Linear Networks
 - ✓ Ring Networks
 - ✓ Mesh Networks.
- UPSR
- BLSR
- } Point to Point
- } Multipoint

⇒ Types of Stations.

- ✓ No-Transition Mobility.
- ✓ BSS - Transition Mobility.
- ✓ ESS - Transition Mobility.

⇒ Telephone companies provide two types of services.

* Analog Services

✓ Analog Switched Services

✓ Analog Leased Services

* Digital Services.

✓ Switched /56 Service

✓ Digital Data Service

⇒ SONET Architecture

* Signals.

✓ Synchronous Transport Signals (STS)

✓ Optical Carriers (OCs)

✓ Synchronous Transport Module (STM)

* SONET Devices.

✓ STS MUX / Demux

✓ Regenerators

✓ Add / Drop Multiplexers and Terminals

* Connections.

✓ Section

✓ Line

✓ Path

- ⇒ IEEE 802.11 defines two MAC Subtypes
- ✓ The Distributed Coordination Function (DCF)
 - ✓ The Point Coordination Function (PCF)

⇒ Frames Types.

- ✓ Management Frames
- ✓ Control Frames
- ✓ Data Frames

⇒ Bluetooth defines two types of Networks

- ✓ Piconet
- ✓ Scatternet

⇒ Three kinds of Connecting Devices.

- ✓ Hubs
- ✓ Multi-layer Switches.
- ✓ Routers.

⇒ Stations are configured in one of three ways:

- ✓ Manually
- ✓ Semi-Automatically
- ✓ Automatically

A Channelization Protocols.

- ✓ FDMA (Frequency Division Multiple Access)
- ✓ TDMA (Time Division " ")
- ✓ CDMA (Code " " ")

- ✓ FDMA (Frequency Division Multiple Access)
- ✓ TDMA (Time Division " " " ")
- ✓ CDMA (Code " " " ")

⇒ Ethernet Evolution. has gone through Four generation.

- * Standard Ethernet (10 Mbps)
- * Fast Ethernet (100 Mbps)
- * Gigabit Ethernet (1 Gbps)
- * 10 Gigabit Ethernet (10 Gbps)

⇒ Telephone network is made up of three major components:

- * Local loops.
- * Trunks
- * Switching offices.

⇒ Telephone network has several levels of switching offices.

- * End offices.
- * Tandem offices.
- * Regional offices.